02. Dataset Understanding

A precise description of the supplied SmartCart dataset and its limitations.

Project: SmartCart Customer Segmentation

Source checked: Sai-Likhith-2005/Customer-Segmentation | smartcart.ipynb | smartcart_customers.csv

Dataset snapshot

Item	Observed
Rows / columns	2,240 rows × 22 columns
Missingness	Income: 24 missing values; all other columns: 0
After filtering	2,236 rows × 15 cleaned columns
After one-hot encoding	2,236 rows × 18 features
Source file	smartcart_customers.csv

Field groups

Group	Fields	Meaning
Identity / time	ID, Year_Birth, Dt_Customer	Customer identifier, birth year, joining date
Demographics / household	Education, Marital_Status, Kidhome, Teenhome	Education, living arrangement and children at home
Economic	Income	Reported customer income
Purchase behaviour	Recency, six Mnt_* fields	Days since last purchase and spend by product category
Channel behaviour	NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, NumStorePurchases, NumWebVisitsMonth	Purchase counts and web visits
Campaign / service	Complain, Response	Complaint flag and campaign response indicator

Important target-like field: Response

Response is a campaign-response indicator. The notebook retains it as one of the clustering input features. If the business goal is to segment customers before a future campaign, this is potentially target leakage: a response observed after a campaign would not be available at segmentation time. Whether it is leakage depends on the timing and intended use, which the repository does not document.

Dataset provenance and caveats

- The archive does not include a data dictionary explaining units, income currency, exact reference period or campaign timing.
- Dt_Customer is parsed with dayfirst=True; the reference date is the maximum date in this dataset.
- The notebook computes Age as 2026 - Year_Birth. That value changes with the year and should be replaced by age at a fixed, documented analysis date or by an age band appropriate to the business question.
- The dataset has no customer-level time series or transaction-level history; purchase counts and spending are pre-aggregated.

Questions before business use

- Are the 2,240 records unique customers?

- Is Income annual household income, individual income, or another measure?
- What currency and price period do the Mnt_* spending fields represent?
- When was Response measured relative to the proposed segmentation and campaign?
- Does the dataset represent the current customer base or a historical cohort?